Department of Biotechnology, University of Tehran

Project 1

Germline Variant Discovery and HLA Analysis from Whole-Exome Sequencing Data

Background:

The Human Leukocyte Antigen (HLA) system represents one of the most polymorphic regions in the human genome and plays a crucial role in immune system function. HLA genes encode proteins responsible for presenting peptides to immune cells, shaping immune recognition and influencing disease susceptibility, drug response, and transplantation outcomes.

Germline variants within HLA loci contribute to the immense diversity of the human immune response and can have functional or clinical implications. By analyzing whole-exome sequencing (WES) data, it is possible to identify inherited variants in these genes and explore their potential relevance to immune-related phenotypes or disease predisposition.

In this project, you will analyze WES data from a human blood sample to identify high-confidence germline variants and characterize HLA gene diversity. The workflow involves downloading and processing sequencing data, performing quality control and alignment, calling germline variants, and finally focusing on HLA loci to interpret their functional and biological significance.

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1. Data Acquisition and Quality Control (QC)

Before any alignment or variant calling, you will assess and clean the raw sequencing data to ensure that your data is of sufficient quality for downstream analyses.

Tasks:

1. Download and SRA-to-FASTQ Conversion:
   - Retrieve the WES dataset in SRA format and convert it to FASTQ files using the SRA Toolkit (fastq-dump or fasterq-dump).
   - Verify file integrity and record the number of reads per file.

2. Initial QC:
    - Perform read quality assessment using FastQC and review the generated HTML reports to evaluate overall sequencing quality.
    - Generate visual summaries of per-base quality scores, GC content, and adapter contamination.
    - Identify potential issues in the sequencing data and discuss their implications for downstream analysis.
3. Read Trimming and Cleaning:
    - Use Trimmomatic to trim low-quality bases and remove adapter sequences.
    - Compare quality reports before and after trimming.
    - Discuss how trimming improves read quality and why it is important for accurate variant calling.
4. Post-QC Evaluation:
    - Re-run FastQC to confirm improvement in read quality.
    - Summarize your QC process in a table showing before/after statistics (e.g., total reads, mean quality, GC%).

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2. Read Alignment and Post-Processing

Once high-quality reads are obtained, align them to the human reference genome and prepare the data for variant calling analysis.

Tasks:

1. Alignment:
    - Align reads to the hg38 reference genome using BWA-MEM.
    - Convert SAM to BAM format, sort, and index the aligned reads.
2. Alignment Evaluation:
    - Assess alignment quality using samtools flagstat.
    - Report metrics such as mapping rate, duplication rate, coverage, and properly paired reads.
    - Discuss how these metrics influence the accuracy and reliability of germline variant calling.
3. Post-Processing:
    - Mark duplicate reads using Picard to remove PCR artifacts and improve variant calling accuracy.
    - Generate a summary of final alignment statistics.

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3. Variant Calling and Filtering

Next, you will identify inherited variants (SNPs and indels) from the aligned reads and filter for high-confidence calls.

Tasks:

1. Variant Calling:
    - Use GATK HaplotypeCaller to identify germline SNPs and indels from the aligned reads.
    - Generate a raw VCF file containing all candidate variants.
2. Variant Filtering:
    - Apply filters based on depth, quality scores, and mapping quality to retain high-confidence variants.
    - Summarize filtered variants (total number, SNP vs indel ratio, transition/transversion ratio, functional categories).
    - Discuss how filtering criteria affect the reliability and interpretability of the results.

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4. HLA Variant Analysis and Interpretation

Finally, you will focus on HLA genes and determine allele composition and germline variants within these loci.

Tasks:

1. HLA Region Extraction:
    - Filter your VCF to retain only variants located in HLA genes (e.g., HLA-A, HLA-B, HLA-C, HLA-DRB1, HLA-DQB1).
    - Use a BED file of HLA gene coordinates and bcftools to find related variants.
    - Verify the number of variants in each gene and summarize basic statistics (e.g., SNP vs. indel ratio, Ti/Tv ratio).
2. HLA Variant Annotation:
    - Annotate the filtered variants using ANNOVAR

    - Collect information on gene name, transcript, predicted consequence (e.g., missense, synonymous, nonsense), and any known clinical associations.
    - Summarize the annotations in a table for each HLA gene.
3. Biological Interpretation:
    - Examine annotated variants to identify those that may influence HLA function or expression, such as variants within peptide-binding regions or rare alleles.
    - Compare your findings with population frequency databases to discuss allele diversity and population relevance.
    - Present your results in tables and figures summarizing variant counts, predicted effects, and possible biological or clinical implications.

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5. Optional: Exploration of Glioma-Associated Variants

Gliomas are malignant brain tumors that arise from glial cells and represent the most common primary brain cancers in adults. Although glioma is primarily driven by somatic mutations, some germline variants can increase disease risk or influence susceptibility. Since the analyzed data originates from the blood of a glioma patient, this step encourages you to reflect on whether any of the identified inherited variants might contribute to glioma predisposition or affect pathways involved in DNA repair, immune regulation, or tumor suppression.

Tasks:

1. Glioma Variant Annotation:
    - Annotate filtered variants using ANNOVAR.
    - Collect information on gene name, transcript, predicted consequence, ClinVar clinical significance, and in silico functional predictions.
2. Biological Interpretation:
    - Examine annotated variants for relevance to glioma predisposition, DNA repair pathways, tumor suppressor genes, or other cancer-associated mechanisms.
    - Highlight variants with strong biological or clinical support and summarize the evidence from literature and public databases.

Keynotes:

The required data can be downloaded from here.

You are required to create a comprehensive Jupyter Notebook report that includes your code, results, and answers to the provided questions. The report should also contain clear Markdown explanations, brief interpretations of your code and analyses, and visual summaries such as FastQC plots, coverage plots, and variant distribution charts. Send your notebook and results as a zip file via email to Mahdi.Anvari7@ut.ac.ir by **November 21, 2025**. Make sure to include your name and student number in the subject line of the email.

Good luck!